

Lab 6: Dimension Reduction

We will use the `Hitters` data set in the `ISLR` package to predict `Salary` for baseball players.

```
library(ISLR)
library(tidyverse)
library(knitr)

str(Hitters)
```

```
## 'data.frame':   322 obs. of  20 variables:
## $ AtBat      : int  293 315 479 496 321 594 185 298 323 401 ...
## $ Hits       : int  66 81 130 141 87 169 37 73 81 92 ...
## $ HmRun      : int   1 7 18 20 10 4 1 0 6 17 ...
## $ Runs       : int  30 24 66 65 39 74 23 24 26 49 ...
## $ RBI        : int  29 38 72 78 42 51 8 24 32 66 ...
## $ Walks      : int  14 39 76 37 30 35 21 7 8 65 ...
## $ Years      : int   1 14 3 11 2 11 2 3 2 13 ...
## $ CAtBat     : int  293 3449 1624 5628 396 4408 214 509 341 5206
## ...
## $ CHits      : int  66 835 457 1575 101 1133 42 108 86 1332 ...
## $ CHmRun     : int   1 69 63 225 12 19 1 0 6 253 ...
## $ CRuns      : int  30 321 224 828 48 501 30 41 32 784 ...
## $ CRBI       : int  29 414 266 838 46 336 9 37 34 890 ...
## $ CWalks     : int  14 375 263 354 33 194 24 12 8 866 ...
## $ League     : Factor w/ 2 levels "A","N": 1 2 1 2 2 1 2 1 2 1 ...
## $ Division   : Factor w/ 2 levels "E","W": 1 2 2 1 1 2 1 2 2 1 ...
## $ PutOuts    : int  446 632 880 200 805 282 76 121 143 0 ...
## $ Assists    : int   33 43 82 11 40 421 127 283 290 0 ...
## $ Errors     : int   20 10 14 3 4 25 7 9 19 0 ...
## $ Salary     : num   NA 475 480 500 91.5 750 70 100 75 1100 ...
## $ NewLeague  : Factor w/ 2 levels "A","N": 1 2 1 2 2 1 1 1 2 1 ...
```

1 Data Processing

1. Remove records with missing data. Create a new (complete) version of your data set. (Hint: `drop_na` in `tidyr` could be helpful.)

2. You may need to create dummy variables for categorical variables in your recipes.
`step_dummy(all_nominal_predictors())` is a good way to do this.
3. You may need to standardize all variables in your recipes.
`step_normalize(all_predictors())` is a good way to do this.

2 Principal Components Regression

The function call `step_pca(all_predictors(), num_comp = d)` will compute d principal components of predictors in a data frame as a component in a recipe.

1. Fit the PCR model using 10-fold cross validation for values of M . Be sure to normalize your predictors.
2. Create a plot of the CV MSE vs. M .
3. When does the smallest cross-validation error occur? Which M would you choose for your final model?
4. How much variability in Y is explained for your chosen value of M ?

3 Partial Least Squares

The function call `step_pls(all_predictors(), num_comp = d)` will compute d partial least squares components of predictors in a data frame as a component in a recipe.

1. Fit the PLS model using 10-fold cross validation for values of M . Be sure to normalize your predictors.
2. Create a plot of the CV MSE vs. M .
3. When does the smallest cross-validation error occur? Which M would you choose for your final model?
4. How much variability in Y is explained for your chosen value of M ?
5. Discuss the difference between PCR and PLS results. Which would you prefer?